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LABORATORIO DE FÍSICA 1
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Instrucciones

Resuelva cuidadosamente cada problema. Si sudas, es normal. Si lloras, también. Asegúrate de incluir todos los pasos, unidades, diagramas (si aplica), y una pizca de paciencia. Si algo no cuadra... probablemente sea culpa del universo (o de las unidades mal convertidas).

Fuente: Física Universitaria, Sears y Zemansky, 13ª Edición, Vol. 1 (2013), Capítulo 12, pág. 394

Problema 1

Enunciado:

Escribe aquí el enunciado del problema (o una versión resumida si lo tienes en el libro).

Solución:

Aquí va la resolución paso a paso del problema.

- Paso 1: Identificar qué demonios está pasando.
- Paso 2: Aplicar la ecuación correspondiente.
- Paso 3: Resolver, mantener la calma, y verificar unidades.
- Paso 4: Escribir la respuesta final con sus respectivas unidades.

Problema 2

Enunciado:

Escribe aquí el enunciado del problema (o una versión resumida si lo tienes en el libro). [1].[2] **Solución:**

$$x_{n+1} = x_n - m f(x) \left[\frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})} \right] \quad (1)$$

For $m = \frac{1+\sqrt{5}}{2} = 1.618$

Steps	xa	xb	xi	f(xi)	(xn+1-xn)
0	1.500000000000	1.600000000000	1.633191538329	-0.001945949763	0.033191538329
1	1.600000000000	1.633191538329	1.564416498720	-0.000020351034	0.068775039609
2	1.633191538329	1.564416498720	1.563240412447	-0.000028545785	0.001176086273
3	1.564416498720	1.563240412447	1.569869184045	-0.000000429797	0.006628771598
4	1.563240412447	1.569869184045	1.570033141259	-0.000000291226	0.000163957214
5	1.569869184045	1.570033141259	1.570590682342	-0.000000021145	0.000557541083
6	1.570033141259	1.570590682342	1.570661309883	-0.000000009115	0.000070627541
7	1.570590682342	1.570661309883	1.570747894571	-0.000000001173	0.000086584688
8	1.570661309883	1.570747894571	1.570768583633	-0.000000000385	0.000020689062

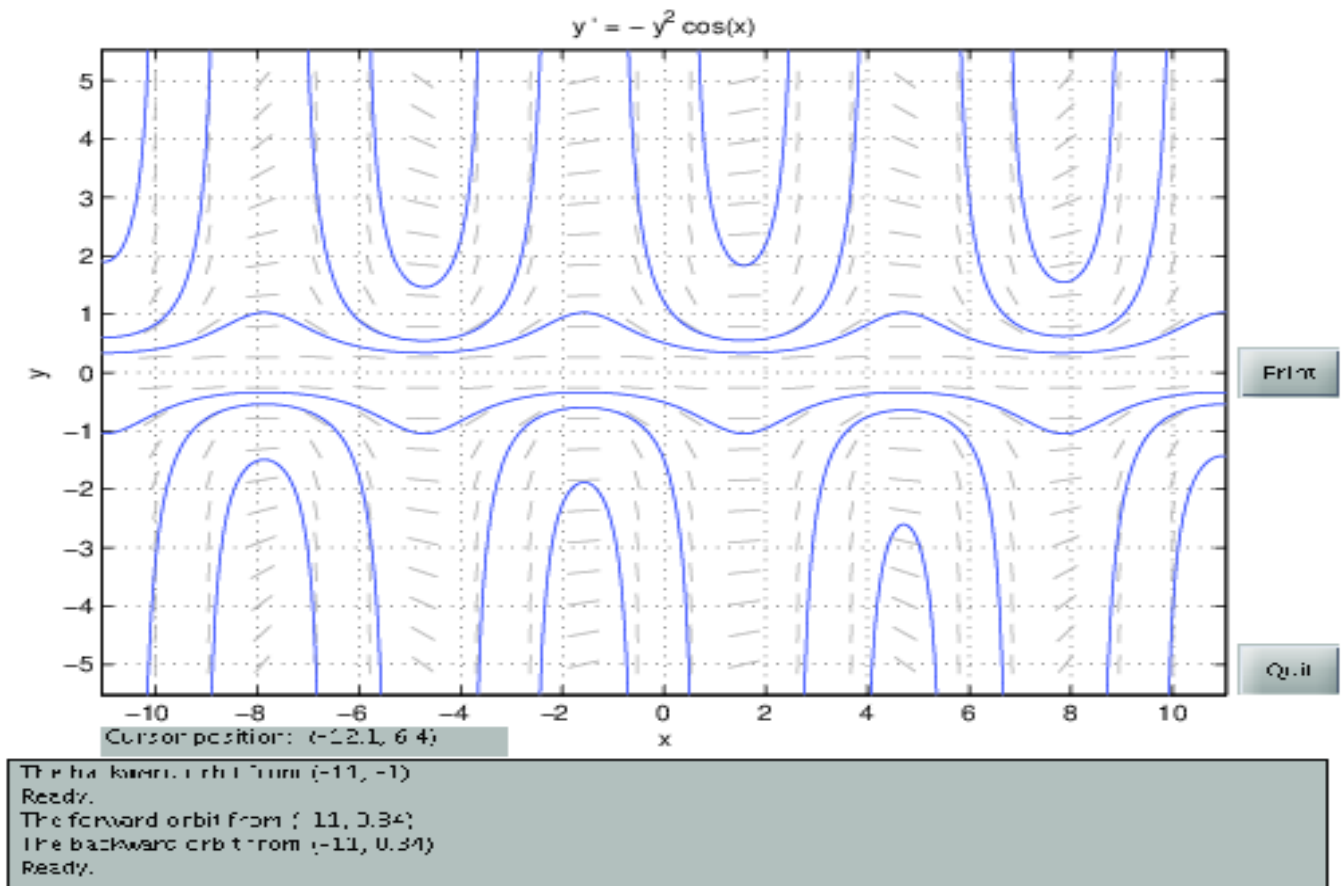


Figure 1: Diagrama esquemático del problema.

9	1.570747894571	1.570768583633	1.570784932394	-0.000000000065	0.000016348761
10	1.570768583633	1.570784932394	1.570790299953	-0.000000000018	0.000005367559
11	1.570784932394	1.570790299953	1.570793673498	-0.000000000004	0.000003373545
12	1.570790299953	1.570793673498	1.570794985787	-0.000000000001	0.000001312289
13	1.570793673498	1.570794985787	1.570795714281	-0.000000000000	0.000000728494

```

1  ! Código Fortran aqui
2  PROGRAM ejemplo
3      IMPLICIT NONE
4      INTEGER :: i, n
5      REAL :: x, sum
6
7      n = 100
8      sum = 0.0
9
10     DO i = 1, n
11         x = REAL(i)
12         sum = sum + x**2
13     END DO
14
15     PRINT *, 'La suma de los cuadrados es: ', sum
16
17 END PROGRAM ejemplo

```

Appendices

A Front-end and Back-end

In software engineering, the terms **front-end** and **back-end** refer to the separation of concerns between the presentation layer (**front-end**), and the data access layer (**back-end**) of a piece of software, or the physical infrastructure or hardware. Therefore in a HPC,

- The login servers are called **front-ends** because you do not run your calculations there.
- Rather run your calculations on **back-end** compute servers.
- The **front-end** server provides access to compute servers via the **batch** system, using the **qsub** command.

B Hard Link or Symbolic Link

The **ln** command is a **Linux/Unix** command used to create file links to an existing file.

B.1 Link types

- There are two types of links
 1. **hard links:** Refer to the specific location of physical data. A hard link allows multiple filenames to be associated with the same file since a hard link points to the inode of a given file, the data of which is stored on disk.
 2. **symbolic links:** Refer to a symbolic path indicating the abstract location of another file. A symbolic links are special files that refer to other files by name.
- The **ln** command by default creates hard links, and when called with the command line parameter **ln -s** creates symbolic links.
- Most operating systems prevent hard links to directories from being created since such a capability could disrupt the structure of a file system and interfere with the operation of other utilities.
- The **ln** command can however be used to create symbolic links to non-existent files.

B.2 Examples

1. Example 1

```
$ ln -s source_file target_file
```

```
$ ls -l source_file target_file
-rw-r--r-- 1 veryv wheel 0 Mar 7 22:01 source_file
lrwxr-xr-x 1 veryv wheel 5 Mar 7 22:01 target_file -> source_file
```

2. Example 2 - Create a symbolic link for /home/Desktop/Links/Example/example.cpp as /home/Test/example.cpp, copy paste the following command

```
$ ln -s /home/Desktop/Links/Example/example.cpp /home/Test/example.cpp
```

```
$ ll
lrwxrwxrwx 1 vivek vivek 16 2007-09-25 22:53 example.cpp -> /home/Desktop/Links/
Example/example.cpp
```

C Securely Copy (SCP) Files

SCP allows files to be copied to, from, or between different hosts (between a local host and a remote host or between two remote hosts.). It uses **ssh** for data transfer and provides the same authentication and same level of security as **ssh**.

1. Copy the file **foobar.txt** from a remote host to the local host

```
$ scp your_username@remotehost.edu:foobar.txt /some/local/directory
```

2. How to **scp** a file to LANL-IC turquoise/

```
$ scp filename username@wtrw.lanl.gov:username@gr-fe.lanl.gov:/remote/path/to/file
```

3. Also shorter probably works

```
scp filename wtrw:gr-fe:/remote/path/to/file
```

References

- [1] N. Ding and S. Williams. An instruction roofline model for gpus. In *2019 IEEE/ACM Performance Modeling, Benchmarking and Simulation of High Performance Computer Systems (PMBS)*, pages 7–18, 2019.
- [2] Charlene Yang, T. Kurth, and S. Williams. Hierarchical roofline analysis for gpus: Accelerating performance optimization for the nersc-9 perlmutter system. *Concurrency and Computation: Practice and Experience*, 32, 2020.